Fariba Khosravani

• 0009-0007-3223-8506 • GitHub: faribakhosravani • LinkedIn: faribakhosravani •
faribakhosravani.khu.ac@gmail.com

Education

Kharazmi University Tehran, Iran
B.Sc. in Geology, GPA 17.12/20 (First-class honours) 2023 – 2027 (expected)

- Executive Vice President, Geological Society of Kharazmi University (June 2025 – Present).
- Editorial Board Member, *Zamin Pouya* Scientific Journal, Vol. 31 (August 2025 – June 2026).

Institute for Advanced Studies in Basic Sciences (IASBS) Zanjan, Iran
Selected participant, 10th School of Earth Sciences Summer 2026

Experience

FEZtool Tehran, Iran
Geographic Information System Specialist November 2023 – Present

- Co-developed `FEZrs`, an open-source Python package for geospatial and multispectral image processing, with 120,000+ downloads.
- Implemented and validated remote-sensing workflows (spectral indices, band math, raster preprocessing) and authored documentation and reproducible usage examples.

Kharazmi University, Rock Mechanics Laboratory Tehran, Iran
Research Assistant (Supervisor: Prof. Mehdi Talkhablou) October 2025 – January 2026

- Tested rock core specimens for Uniaxial Compressive Strength (UCS) following ISRM and ASTM D7012, controlling L/D ratio and end-face tolerances.
- Processed stress–strain records to derive strength and deformability parameters, and reported results with quality-control checks on specimen preparation.

Publications

Journal Articles (Under Review)

[1] Farmahinifarahani, M., Bagheri, M., & **Khosravani, F.** SeisFAV-Net: An integrated network combining Fourier Neural Operator, self-attention mechanisms, and variational autoencoder for seismic data random noise attenuation. *Applied Computing and Geosciences* [under review].

[2] Farmahinifarahani, M., Elmi, P., Nedaee, M., Kiani Faizabadi, M., Mirzaee, H., Moradi, P., Yazdanifar, M., **Khosravani, F.**, Najafi, F., & Talkhablou, M. FEZrs: An open-source Python package for geospatial multispectral image processing and remote sensing. *Journal of Open Source Software* [under review].

Conference Proceedings

[3] **Khosravani, F.**, Farmahinifarahani, M., & Bagheri, M. (2026). Fourier Neural Operators with pointwise Convolutional Neural Networks for seismic data random noise attenuation. *22nd Iranian Geophysical Conference (IGC22).*

Projects

GEMCNN June 2025 – May 2026
github.com/faribakhosravani/GEMCNN

- Deep-learning pipeline for gemstone origin determination and treatment detection from multimodal analytical data (FTIR, XRF, ICP-MS).
- Built end-to-end training and evaluation workflow with reproducible configuration and documented inference scripts.

Fault_Monitoring October 2025 – June 2026
github.com/faribakhosravani/Fault_Monitoring

- Automated seismicity monitoring and visualization for major active fault systems in Iran, using public earthquake catalogues.

A Comprehensive Review of Cenozoic Tectonic Events in Iran March 2026 – July 2026

- Capstone project for the Geology of Iran course: synthesis of Cenozoic tectonic evolution from published structural and geochronological data.

Workshops & Certifications

- Seismology Skill Building Workshop 2026 (*EarthScope*, in progress).
- Applied Geotechnical Seismology: Design, Implementation, Data Processing and Analysis, theoretical and practical (*Kharazmi University*).
- Non-Destructive Geophysical Methods in Engineering Geology: GPR, Seismology and Magnetometry, theoretical and practical (*Kharazmi University*).
- Hydrocarbon Reservoir Management Series: rock typing for field development, and micro-/nanoscale reservoir characterization from imaging experiments (*Kharazmi University*).
- Gemstone Cutting: faceting, semi-faceting, cabochon (*Technical and Vocational Training Organization*).

Skills & Interests

Programming: Python (NumPy, pandas, matplotlib), Linux/Bash (familiar)
Software: ArcGIS, Google Earth Engine, MODFLOW, CorelDRAW
Laboratory: Rock mechanics (UCS, specimen preparation), XRF, FTIR, ICP-MS
Languages: Persian (native), English (C1, preparing for TOEFL), Spanish (A2)
Interests: Tectonics, Seismology, Structural Geology, Rock Mechanics